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**Course:** Calculus Beta

**Assignment:** MyLab Math - Uge 2

- a. Find the slope of  $x^3 + y^3 - 217xy = 0$  at the points (6,36) and (36,6).
- b. At what point other than the origin does the curve have a horizontal tangent line?
- c. Find the coordinates of the point other than the origin where the curve has a vertical tangent line.

a. Perform implicit differentiation to find an equation for the slope of the curve. First, differentiate both sides of the equation with respect to  $x$ , treating  $y$  as a differentiable function of  $x$ . Then, collect the  $\frac{dy}{dx}$  terms on one side and solve for  $\frac{dy}{dx}$ .

Differentiate  $x^3 + y^3 - 217xy = 0$  with respect to  $x$ .

$$3x^2 + 3y^2 \frac{dy}{dx} - 217y - 217x \frac{dy}{dx} = 0$$

Collect the  $\frac{dy}{dx}$  terms on one side and solve for  $\frac{dy}{dx}$ .

$$\begin{aligned} \frac{dy}{dx} (3y^2 - 217x) &= 217y - 3x^2 \\ \frac{dy}{dx} &= \frac{217y - 3x^2}{3y^2 - 217x} \end{aligned}$$

Finally, substitute the coordinates of a point into the equation for  $\frac{dy}{dx}$  and simplify to find the slope of the curve at that point.

Substitute the point (6,36) into the equation for  $\frac{dy}{dx}$ .

$$\frac{dy}{dx} = \frac{217(36) - 3(6)^2}{3(36)^2 - 217(6)}$$

Simplify to find the slope of the curve  $x^3 + y^3 - 217xy = 0$  at (6,36).

The slope is  $\frac{1284}{431}$ .

Now substitute the point (36,6) into the equation for  $\frac{dy}{dx}$ .

$$\frac{dy}{dx} = \frac{217(6) - 3(36)^2}{3(6)^2 - 217(36)}$$

Simplify to find the slope of the curve  $x^3 + y^3 - 217xy = 0$  at (36,6).

The slope is  $\frac{431}{1284}$ .

b. A curve has a horizontal tangent line at a point if the derivative of the curve at that point equals zero. Since the derivative is a fraction, it will equal zero when the numerator equals zero.

$$217y - 3x^2 = 0$$

Solving this equation for one variable in terms of the other will result in an expression that can be substituted back into the original equation to find the point in question. Since one of the variables is being squared and the other is not, it is advisable to solve for the variable that is not being squared in terms of the one that is.

Solve for  $y$  in terms of  $x$ .

$$y = \frac{3x^2}{217}$$

Substitute this expression for  $y$  in the original equation.

$$x^3 + \left(\frac{3x^2}{217}\right)^3 - 217x\left(\frac{3x^2}{217}\right) = 0$$

Simplify and factor the equation.

$$x^3 + \left(\frac{3x^2}{217}\right)^3 - 217x\left(\frac{3x^2}{217}\right) = 0$$

$$x^3 + \frac{3^3 x^6}{217^3} - 3x^3 = 0$$

$$x^3 \left( \frac{3^3 x^3}{217^3} \right) - x^3(2) = 0$$

$$x^3 \left( \frac{3^3 x^3}{217^3} - 2 \right) = 0$$

Now apply the zero-factor property and solve for x.

$$\begin{aligned} x^3 &= 0 & \frac{3^3 x^3}{217^3} - 2 &= 0 \\ x &= 0 \end{aligned}$$

$$\frac{3^3 x^3}{217^3} = 2$$

$$x^3 = \frac{217^3 \cdot 2}{3^3}$$

$$x = \frac{217}{3} \sqrt[3]{2}$$

Note that if  $x = 0$ , then  $y = 0$ , because the  $x$ - and  $y$ -values must satisfy the equation  $x^3 + y^3 - 217xy = 0$ , so to find the point other than the origin, determine the value of  $y$  for the other solution. Substitute  $\frac{217}{3} \sqrt[3]{2}$  for  $x$  in the equation  $y = \frac{3x^2}{217}$  and calculate  $y$ .

$$\begin{aligned} y &= \frac{3 \left( \frac{217}{3} \sqrt[3]{2} \right)^2}{217} \\ &= \frac{217}{3} \sqrt[3]{2^2} \\ &= \frac{217}{3} \sqrt[3]{4} \end{aligned}$$

Therefore, the coordinates of the point other than the origin where the curve has a horizontal tangent line are  $\left( \frac{217}{3} \sqrt[3]{2}, \frac{217}{3} \sqrt[3]{4} \right)$ .

c. A curve has a vertical tangent line at a point if the derivative of the curve at that point is undefined. Since the derivative is a fraction, it will be undefined when the denominator equals zero.

$$3y^2 - 217x = 0$$

As before, solving this equation for one variable in terms of the other will result in an expression that can be substituted back into the original equation to find the point in question. Since one of the variables is being squared and the other is not, it is advisable to solve for the variable that is not being squared in terms of the one that is.

Solve for  $x$  in terms of  $y$ .

$$x = \frac{3y^2}{217}$$

Substitute this expression for  $x$  in the original equation.

$$\left( \frac{3y^2}{217} \right)^3 + y^3 - 217 \left( \frac{3y^2}{217} \right) y = 0$$

Simplify and solve the equation for  $y$ . Rather than go through more arduous algebraic simplifications, however, notice that this equation is nearly identical to the equation that resulted in the previous problem when an expression containing  $x$  was substituted in for  $y$ . The only difference, in fact, is that all the variables that were  $x$ 's before are now  $y$ 's. Therefore, all of the algebra will be the same, and the answer will be the same, merely applying to  $y$  instead of  $x$ .

$$y = \frac{217}{3} \sqrt[3]{2}$$

Substitute this value for  $y$  in the expression where you solved for  $x$  in terms of  $y$  and solve for  $x$ .

$$\begin{aligned} x^3 &= \frac{3 \left( \frac{217}{3} \sqrt[3]{2} \right)^2}{217} \\ &= \frac{217}{3} \sqrt[3]{2^2} \\ &= \frac{217}{3} \sqrt[3]{4} \end{aligned}$$

Therefore, the coordinates of the non-origin point where the curve has a vertical tangent line are  $\left( \frac{217}{3} \sqrt[3]{4}, \frac{217}{3} \sqrt[3]{2} \right)$ .